

Unknot Recognition by Structural Certificates and Compressed Chain Complexes

Exact improvements and a finite-type route towards quasi-polynomial time

Research continuation for the ProveIt project

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Abstract

We continue the ProveIt unknot-recognition implementation from a pinned, fully tested source snapshot. Two complementary improvements are developed. First, signed Seifert-graph certificates give a linear-time decision procedure (after validation, in the word-RAM model) on every homogeneous knot diagram and an additional one-sided obstruction on nonhomogeneous diagrams. Second, a scanning Khovanov complex is represented as a sum of distinct, explicitly serialized differential components with multiplicity polynomials. Repeated components are continued once, even when their virtual multiplicity is exponential. A decision-only specialization replaces those polynomials by the four-element semiring of nonnegative counts saturated at three; this preserves the final unreduced rank-two unknot test. An optional exact suffix-Euler calculation supplies monotone lower bounds on completed homology rank, sometimes deciding knottedness before closure.

We prove preservation of the represented complex, the saturation theorem, and a quantitative finite-type bound. If w is the maximum frontier size and b the maximum connected differential-component size produced after elimination, the number of stored component types at any stage is at most $\sum_{s \leq b} s^s ((w-1)!!)^s 2^{s^2 2^{w/2}}$. This yields the specific $n^{O(\log n)}$ target whenever $b^2 2^{w/2} = O(\log^2 n)$, with no restriction on the number of repeated copies. The hypothesis is an execution parameter, not a proved universal property of knot diagrams.

The deliverable includes the implementation, independent certificate replay, regression tests, randomized comparisons, paired timing data, an unchanged reference implementation, and integration instructions. Explicit-basis homology computation can still require exponential time even for three-strand braids. We do not establish a general quasi-polynomial unknot recognizer. The new results isolate a concrete mechanism by which a complete implementation can avoid multiplicity growth, and identify the structural bounds still needed to turn that mechanism into a general complexity theorem.

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1 Results, scope, and source snapshot

1.1 What has been achieved

The objective is to improve the actual recognizer in `Topology/UnknotRecognition/fast`, with the target

$$T(n) = n^{O(\log n)} = 2^{O((\log n)^2)} \quad (1)$$

for a diagram with n crossings. The source snapshot used throughout is commit `4e6fe879e5238ec0b134b6d33fdca0b8c7c1c711` of the ProveIt repository [14]. Forty source, test, example, and configuration files were retrieved and checked against their Git blob identifiers. The baseline’s 41 unit tests pass in this environment. The supplied provenance manifest also records SHA-256 digests of those exact bytes. The reference copy is not an earlier acceleration proposal: it is the current Python implementation at the stated commit.

The contributions have different logical statuses. Keeping those statuses separate is necessary for interpreting both the mathematics and the timing results.

Contribution	Precise scope	Status
Seifert certificates	Genus-zero canonical surfaces, homogeneous diagrams, and a Rasmussen interval excluding zero	Implemented; known topological inputs, proved algorithmic use
Component sharing	Repeated actual differential components, identified by complete equality of serialized data up to homological shift	Implemented; correctness proved here
Saturated decision weights	Counts capped at three, with no absolute grading information in the multiplicities	Implemented; correctness proved here
Suffix-Euler certificates	Componentwise completed Euler characteristics as a monotone homology-rank lower bound	Implemented; correctness proved here
Finite-type complexity	A bound in the actual frontier size w and post-elimination component size b	Proved sufficient condition
General bound (1)	All valid classical knot diagrams	Open for this implementation

The algebraic arguments are presented as research results of this continuation. We do not claim that direct-sum sharing, semiring quotients, or the underlying topological criteria are new concepts. The contribution is their precise, compatible realization for this scanner, the associated complexity accounting, and the tested implementation. No literature-wide priority claim is made for the general compression principles.

1.2 What the existing code already did

The pinned recognizer validates a one-component spherical PD diagram; applies incremental Reidemeister I/II reduction; tests whether a descending traversal exists; and splits visible connected sums. Each remaining factor is tested using modular Alexander and Jones obstructions. The exact Alexander polynomial is optional when the modular test has already run. A budgeted Reidemeister III search can unlock further reductions. The complete fallback computes Khovanov ranks over $\mathbb{F}_2$ by a Bar-Natan scan [4].

This code already includes bit-packed cobordism coefficients, compiled composition and transfer plans, Markowitz pivot selection, shape caches, multiple scan orders, and optional races. Reimplementing those changes would not advance the current project. The new structural stage is placed before factorization and the matrix-based filters. The new compressed backends are optional,

because their bookkeeping can cost time when few equal components occur. The ordinary optimized scanner remains the default fallback.

1.3 Current relation to the hierarchy programme

Lackenby’s 2026 hierarchy preprint gives a constructive topological route and certification results. Its Proposition 9.1 bounds iterations in terms of hierarchy length L and pattern complexity g by $L(g+1)^L$; Proposition 10.2 gives $L \leq 4c_q(\mathcal{H})$ for a refined construction. The paper does not supply the end-to-end quasi-polynomial implementation bound needed here. The 2021 author slides announce a $2^{O((\log n)^3)}$ bound, which is quasi-polynomial but is a different exponent from (1) [10, 9].

Our changes do not implement those missing geometric operations. They improve the existing exact algebraic route and supply a different, explicitly parameterized compression theorem. This distinction prevents an iteration bound, a fast special case, or a compressed data structure from being mistaken for a complete general complexity proof.

2 Input model and the exact rank threshold

2.1 Diagrams, arithmetic, and promises

A nonempty PD diagram has n ordered crossing rows. A row (a, b, c, d) lists four counterclockwise ports, with the under-strand joining ports 0, 2 and the over-strand joining 1, 3. Every edge label occurs exactly twice. Validation checks the single component and the spherical rotation system. The empty PD code denotes one crossing-free circle, not the empty link. Braid and grid input are converted to this representation.

The low-level scan routines, like the inherited routines, take a PD array under a validity promise. The public command line obtains it through `Diagram.from_json`. An arbitrary direct call to the permissive `Diagram` constructor does not constitute validation. Every theorem asserting an unknot verdict in this report assumes a valid classical one-component diagram.

The parameter n refers to an explicitly listed diagram. After label normalization, its encoding uses $O(n \log(n+2))$ bits. An input containing very long integer labels necessarily incurs the cost of reading and normalizing those labels. Linear graph algorithms below use a word-RAM with $\Theta(\log(n+2))$ -bit words; their bit costs gain the usual polynomial logarithmic factors. Homological multiplicities use arbitrary-precision nonnegative integers; morphisms use exact $\mathbb{F}_2$ coefficients.

2.2 Why unreduced rank two detects the unknot

We use total dimension, with the quantum grading forgotten. Define

$$r_2(K) = \dim_{\mathbb{F}_2} \text{Kh}(K; \mathbb{F}_2), \quad \tilde{r}_2(K) = \dim_{\mathbb{F}_2} \widetilde{\text{Kh}}(K; \mathbb{F}_2).$$

Shumakovitch’s splitting over $\mathbb{F}_2$ gives $r_2(K) = 2\tilde{r}_2(K)$, in fact degree by degree in the homological grading [15, Corollary 3.2.C]. The rational-rank consequence of Kronheimer–Mrowka says that every nontrivial knot has reduced rational Khovanov rank greater than one [8]. The universal coefficient theorem for the finite free integral reduced complex gives

$$\dim_{\mathbb{Q}} \widetilde{\text{Kh}}(K; \mathbb{Q}) \leq \tilde{r}_2(K).$$

Therefore

$$K \text{ is the unknot} \iff \tilde{r}_2(K) = 1 \iff r_2(K) = 2. \quad (2)$$

For the forward implication, the reduced unknot complex has one generator. For the reverse implication, rational rank greater than one is impossible. One must not substitute the incorrect argument that dimension one over $\mathbb{F}_2$ automatically excludes odd integral torsion. The rational-rank theorem is what makes the coefficient change sufficient.

The normalized homological degree is obtained from cube degree by a uniform writhe-dependent shift. The existing scanner reports raw cube degrees. Uniform shifts do not affect total rank, and the new exact-rank backend preserves the same raw degree convention. No quantum-graded homology or integral torsion is claimed by the delivered API.

3 A linear structural decision stage

3.1 Signed Seifert graphs

Apply the oriented smoothing at every crossing of an oriented diagram D . Let $s = s(D)$ be the number of resulting Seifert circles. The signed Seifert graph $G(D)$ has these circles as vertices and one signed edge for each crossing. Let G_+ and G_- be its two spanning subgraphs obtained by keeping only the indicated sign, and put

$$c_+ = \#\pi_0(G_+), \quad c_- = \#\pi_0(G_-), \quad \delta(D) = s + 1 - c_+ - c_-.$$

Isolated vertices are counted in *both* spanning subgraphs. Omitting them invalidates the formulas even on small examples.

The canonical Seifert surface has s discs and n bands. Because a knot diagram is connected, that surface is connected and has one boundary component. Consequently its genus is

$$g_D = \frac{n - s + 1}{2}. \quad (3)$$

This is an upper bound on knot genus for any diagram. Equality need not hold without an additional hypothesis.

Lemma 3.1 (A graph-theoretic interpretation of the defect). *Let G be a loopless connected finite multigraph whose edges have signs $+$ or $-$, including possible parallel edges. Let s be its number of vertices and $c_{\pm}$ the component counts of its signed spanning subgraphs. Then $\delta = s + 1 - c_+ - c_- \geq 0$. Moreover, $\delta = 0$ if and only if every block of G is monochromatic, equivalently, every simple cycle is monochromatic.*

Proof. Write m, m_+, m_- for the edge counts. The $\mathbb{F}_2$ cycle spaces satisfy $Z(G_+) \oplus Z(G_-) \subseteq Z(G)$, with dimensions

$$\dim Z(G) = m - s + 1, \quad \dim Z(G_+) = m_+ - s + c_+, \quad \dim Z(G_-) = m_- - s + c_-.$$

Their codimension is precisely $s + 1 - c_+ - c_-$, proving nonnegativity. If a simple cycle contains both signs, its positive edge set has nonzero boundary at a sign-change vertex. It therefore cannot be the sum of a positive cycle and a negative cycle. Thus a mixed simple cycle prevents codimension zero. Conversely, if every simple cycle is monochromatic, the usual generation of the cycle space by simple cycles shows that the two signed cycle spaces span $Z(G)$.

A cycle lies in a block. Conversely, any two edges in a nontrivial 2-connected block lie on a common cycle; a mixed block would therefore contain a mixed cycle. Bridges are one-edge blocks. These observations give the stated block formulation, also for parallel edges, whose two-edge cycles must be included. $\square$

There is another useful realization of the same integer. Form a bipartite graph B with one vertex for each component of G_+ and one for each component of G_- . Every vertex v of G gives an edge joining the two components containing v . The graph B is connected, and

$$\beta_1(B) = s - (c_+ + c_-) + 1 = \delta.$$

This interpretation also explains why a nearly homogeneous graph need not be small: a small cycle rank controls neither the number of edges nor the lengths of its paths.

A diagram is called homogeneous when each block of its signed Seifert graph has a single sign. Lemma 3.1 is the graph mechanism behind Abe’s criterion for homogeneous diagrams [1]. Cromwell’s theorem states that the canonical surface of a homogeneous diagram realizes link genus; we need its knot case [5, 12].

3.2 The three exact certificates

Let w be writhe in a consistent crossing-sign convention. The Kawamura–Lobb bounds give the interval

$$L(D) = w - s + 2c_+ - 1 \leq s_R(K) \leq U(D) = w + s - 2c_- + 1, \quad (4)$$

where s_R denotes Rasmussen’s invariant [11, Theorem 1.10]. Its width is $U - L = 2\delta$. The unknot has $s_R = 0$.

Theorem 3.2 (Structural decision theorem). *For a valid knot diagram, the following rules are sound:*

1. If $g_D = 0$, return UNKNOT.
2. If $g_D > 0$ and $\delta = 0$, return KNOTTED.
3. If $L(D) > 0$ or $U(D) < 0$, return KNOTTED.

If no rule applies, return no certificate. All the necessary quantities can be computed in $O(n)$ word operations and $O(n)$ words of storage after validation. Thus this stage alone decides every homogeneous knot diagram in linear time.

Proof. A connected orientable surface of genus zero with one boundary component is a disc. In the first case, the canonical embedded Seifert surface is such a disc, so its boundary is the unknot. In the second case, homogeneity and Cromwell’s theorem identify g_D with knot genus, which is positive. The third case contradicts the value zero of Rasmussen’s invariant for the unknot, by (4).

For the complexity, orient the knot by one traversal. In a crossing with incoming under-port u and incoming over-port o , oriented smoothing pairs u with $o + 2$ and o with $u + 2$, modulo four. Walking the smoothing involution alternately with edge pairing labels all Seifert circles in $O(n)$ steps. Each crossing now gives one signed edge. Two adjacency-list component traversals compute c_+ and c_- in $O(n + s)$ operations; here $s \leq n + 1$. The arithmetic uses $O(\log(n + 2))$ -bit integers. $\square$

The interval $[0, 0]$ is not an unknot certificate. The figure-eight diagram, for example, is homogeneous with canonical genus one, yet its Rasmussen interval is $[0, 0]$. It is decided by the genus rule.

A sign convention carried over from the source. The repository’s PD sign function and its positive Artin-generator documentation use opposite global conventions on the standard braid fixtures. The delivered code retains the sign function used by the existing filters. A global sign reversal exchanges c_+ with c_- , replaces w by $-w$, and reflects $[L, U]$ to $[-U, -L]$. Every verdict above is invariant under that reflection. Consequently no signed value of s_R in an external convention is asserted by the witness without this qualification; the knottedness certificate is unaffected. The source’s established Jones and scanner conventions are not silently changed.

3.3 Independent replay and placement in the pipeline

A certificate records $n, w, s, c_+, c_-, \delta, g_D, [L, U]$, the deciding rule, and the verdict. The replay function independently rebuilds the edge pairing and knot orientation, computes Seifert circles with union-find, and recomputes the signed component counts. It compares all recorded integers and then checks the indicated sufficient condition. Its cost is $O(n\alpha(n))$ word operations using path compression and union by size. It is an independent implementation of the combinatorial calculation, not a formal proof checker for the imported topological theorems.

The new default calls the certificate on the original validated diagram, before reduction, factorization, or determinants. That location is necessary for the straightforward linear-time theorem for the entire homogeneous family: a superlinear preliminary routine would otherwise dominate it. There is a tradeoff on tiny diagrams. An inconclusive graph calculation adds work to cases the old pipeline settles almost immediately. The paired measurements report this overhead rather than hiding it in an aggregate. The option `-no-seifert` restores the old pipeline’s stage order.

3.4 Braid specialization and a counterexample to an overstrong bound

Proposition 3.3. *Let D be the closure diagram of a t -strand braid word with one component. Then $s(D) = t$. If J_+ and J_- are the sets of Artin-generator indices occurring with the two signs, then*

$$\delta(D) = |J_+ \cap J_-|, \quad g_D = \frac{n - t + 1}{2}.$$

Proof. Oriented smoothing produces one circle per braid position. The underlying Seifert graph is the path on t vertices with parallel crossing edges. A spanning subgraph of that path using the index set J has $t - |J|$ components, irrespective of repetitions. Connectivity of the braid projection gives $|J_+ \cup J_-| = t - 1$. Substitute these component counts into the definition of δ . Equation (3) gives the second formula. $\square$

Example 3.4 (Defect one with arbitrarily large canonical genus). *For $m \geq 2$, consider the two-strand closure of $\sigma_1^m \sigma_1^{-(m-1)}$. The braid reduces to σ_1 , so its closure is an unknot. Its displayed diagram has $2m - 1$ crossings, two Seifert circles, $\delta = 1$, and $g_D = m - 1$. In particular, neither a lower bound $g(K) \geq g_D - \delta$ nor a bound on canonical genus in terms of defect alone can support a general algorithm.*

The example is readily simplified by the existing Reidemeister engine. Its purpose is a mathematical obstruction to an invalid parameter argument, not a claim that these diagrams are difficult inputs.

3.5 The defect obstruction persists after Reidemeister I and II

The preceding cancellation example might suggest first performing all crossing-decreasing Reidemeister I and II moves, and then trying to bound surface complexity by the homogeneity defect. The following explicit family rules out that repair as well.

Proposition 3.5 (Unbounded excess genus at defect one after I/II). *For every $k \geq 2$, the standard three-strand closure diagram E_k of*

$$\beta_k = \sigma_2^k \sigma_1 \sigma_2 \sigma_1^{-k}$$

is an unknot diagram with $2k + 2$ crossings. It has no available crossing-decreasing Reidemeister I or II move, has homogeneity defect $\delta(E_k) = 1$, and has diagram Seifert genus $g_{E_k} = k$.

Proof. The braid relation gives

$$\sigma_2 \sigma_1 \sigma_2 \sigma_1^{-1} = \sigma_1 \sigma_2.$$

Therefore

$$\beta_k = \sigma_2^{k-1} (\sigma_2 \sigma_1 \sigma_2) \sigma_1^{-k} = \sigma_2^{k-1} \sigma_1 \sigma_2 \sigma_1^{-(k-1)} = \beta_{k-1}.$$

Induction gives $\beta_k = \beta_0 = \sigma_1 \sigma_2$, whose closure is an unknot. Each induction step consists of one braid Reidemeister III move and one inverse-pair Reidemeister II cancellation.

The diagram has three Seifert circles. Generator index 1 appears with both signs and index 2 with just one sign. The braid graph formula gives $\delta = 1$, and the surface formula gives $g_{E_k} = ((2k + 2) - 3 + 1)/2 = k$.

For completeness, the assertion about local reductions can be checked uniformly rather than inferred from numerical examples. Number the crossings by $0, \dots, 2k + 1$ in braid order, and number the four counterclockwise ports at each crossing by $0, 1, 2, 3$, with ports $0, 2$ under. A PD code for this closure, on edge labels $0, \dots, 4k + 3$, is

crossing	PD row
0	$(2k + 3, 4k + 3, 0, 1)$
$1 \leq i < k$	$(2i - 1, 2i - 2, 2i, 2i + 1)$
k	$(2k - 2, 4k + 2, 2k, 2k + 1)$
$k + 1$	$(2k - 1, 2k + 1, 2k + 2, 2k + 3)$
$k + 2$	$(2k, 2k + 4, 2k + 5, 2k + 2)$
$k + 3 \leq i \leq 2k + 1$	$(2i - 2, 2i, 2i + 1, 2i - 1).$

Writing (i, j) for a dart, the facial walks are precisely the following. Increasing or decreasing ranges include both endpoints.

length	facial walk
$k + 1$	$(0, 0), (k + 1, 0), (k - 1, 0), \dots, (1, 0)$
$k + 2$	$(0, 1), (2k + 1, 3), (2k, 3), \dots, (k + 1, 3)$
$k + 2$	$(0, 2), (1, 2), \dots, (k - 1, 2), (k, 1), (2k + 1, 2)$
3	$(k - 1, 3), (k + 1, 1), (k, 0)$
$k + 1$	$(k, 2), (k + 2, 1), (k + 3, 1), \dots, (2k + 1, 1)$
3	$(k, 3), (k + 1, 2), (k + 2, 0)$
2	$(i, 3), (i + 1, 1), \quad 0 \leq i \leq k - 2$
2	$(i, 2), (i + 1, 0), \quad k + 2 \leq i \leq 2k.$

Pairing equal edge labels and advancing one counterclockwise port verifies each walk. The listed walks use all $8k + 8$ darts exactly once, so the list is exhaustive. Their number is $2k + 4 = n + 2$, as required for the spherical rotation system.

There is no monogon. Since $k \geq 2$, the only digons are the last two families in the table. Each lies within a run of crossings of the same Artin generator and the same sign. A cancellable oriented Reidemeister II pair has opposite crossing signs. Hence none of these digons can be cancelled, and no I or II move is available. $\square$

The artifact `defect_one_family.py` supplies the closed-form PD rows and facial walks, verifies them against the existing braid converter and face traversal, checks the absence of local reductions, and checks the braid-word identity step. It was run for every $2 \leq k \leq 100$, covering 99 examples with up to 202 crossings. These finite checks validate the implementation of the formulas; the displayed symbolic proof establishes the statement for all k .

This family is not asserted to be difficult for the completed recognizer: the explicit sequence of III and II moves supplies a short reduction, and the existing late-III search can expose it. Its role is to show that δ , even after all immediately available I/II reductions, does not control $g_D - g(K)$, and thus cannot by itself justify a genus-based bound on the scanner's remaining state complexity.

4 Why compressed multiplicity matters

4.1 A basis-size obstruction, not a decision lower bound

Kelomäki–Schütz prove that explicit-basis scanning and divide-and-conquer Khovanov algorithms can take exponential time even on positive or alternating three-strand braids. Their paper also gives a polynomial-time method for the homology of closed three-strand braids using additional structure [7]. Thus large homology dimension obstructs an explicit basis, not a short binary description of its dimensions.

For a concrete illustration, let

$$W_m = (\widehat{\sigma_1 \sigma_2^{-1}})^m, \quad 3 \nmid m.$$

Its permutation is a power of a three-cycle, so the closure is a knot. The diagram has $2m$ crossings, three Seifert circles, and $\delta = 0$. For $m \geq 2$, Theorem 3.2 certifies genus $m - 1 > 0$ and hence knottedness in $O(m)$ operations. The determinant formula is

$$d_m = \lambda^m + \lambda^{-m} - 2, \quad \lambda = \frac{3 + \sqrt{5}}{2}, \quad (5)$$

also expressible as $L_{2m} - 2$ in Lucas numbers [3]. The graded Euler-characteristic identity for reduced Khovanov homology, evaluated at the unit-modulus argument corresponding to $V_K(-1)$, gives $|V_K(-1)| \leq \dim_{\mathbb{F}_2} \widehat{\text{Kh}}(K; \mathbb{F}_2)$ by the triangle inequality. Consequently an explicit unreduced homology basis has at least $2d_m$ elements. This lower bound does not require an additional thinness theorem.

One can avoid irrational arithmetic in this witness family. With

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}, \quad t_0 = 2, \quad t_1 = 3, \quad t_{m+1} = 3t_m - t_{m-1},$$

we have $t_m = \text{tr}(A^m)$ and $d_m = t_m - 2$. The two eigenvalues of A are λ, λ^{-1} . The determinant identity can also be obtained from the reduced Burau closure formula [13]. This supplies a simple exact recurrence for checking all displayed values.

The growth $d_m = \Theta(\lambda^m)$ does not conflict with a linear-time *decision* on this family. The baseline already rejects these knots through its polynomial filters. The new graph stage improves their preprocessing asymptotics; it does not solve an instance family that was previously undecidable by the code.

4.2 Three different notions of size

The scanner's state has at least three independently relevant sizes:

1. the frontier size, which controls the number and algebra of matching objects;
2. the size of a connected component of the nonzero differential graph;
3. the number of independent copies of equal components, possibly in many homological degrees.

An optimized coefficient representation reduces the cost of one morphism. A good crossing order can reduce the frontier. Neither operation by itself prevents an exponential number of equal summands. Conversely, a direct-sum compression does not help a large indecomposable differential component. These distinctions determine both the theorem and the limitations below.

5 Exact component sharing in the scanner

5.1 The algebraic state being represented

We work in the finite $\mathbb{F}_2$ matching-and-dot algebra already used by the validated scanner, with quantum grading forgotten. At a fixed frontier, an object is a matching. A differential matrix connects objects in consecutive homological degrees. Its entries are the complete dot-mask cobordism values; addition is XOR and composition is the existing exact composition operation. The relative correctness theorem assumes that the inherited extension, delooping, and composition routines implement the usual ungraded $\mathbb{F}_2$ Bar–Natan category. The independent tests below support that implementation assumption; they are not a formal verification of the full category.

The differential graph of a complex has one vertex per object and an undirected edge whenever the differential has a nonzero entry in either direction. Its connected components specify literal direct summands. This is a matrix-level decomposition, requiring no search for a basis change. A component may still decompose further after an undiscovered basis change; our method makes no claim to find indecomposable objects.

Definition 5.1 (Template and multiplicity polynomial). *A template T is a finite connected differential component with minimum homological degree normalized to zero, with numbered objects and all matching and differential data retained. For $p(z) = \sum_h p_h z^h \in \mathbb{N}[z]$, define*

$$p \odot T = \bigoplus_h T[h]^{\oplus p_h}.$$

A compressed state is a finite collection (p_a, T_a) representing $\bigoplus_a p_a \odot T_a$.

The polynomial coefficients are ordinary nonnegative integers. They are *not* elements of $\mathbb{F}_2$. For example, two isolated copies of a generator represent a two-dimensional vector space; replacing their multiplicity by $2 \bmod 2 = 0$ would destroy the represented homology.

5.2 Equality by complete serialization

For a connected component, list its object indices in their inherited order and renumber them $0, \dots, s-1$. Subtract the minimum degree h_0 . The key is the ordered tuple

$$(m_v, h_v - h_0, ((u, d_{vu}))_{u: d_{vu} \neq 0})_{v=0}^{s-1}, \tag{6}$$

with each row sorted by its local target index. Matching identifiers refer to one shared **Planar** table, so equality means equality of the actual matching pairs at this stage. All coefficient masks appear in the key.

Equal keys exhibit a chain isomorphism, by mapping the vertex in each local position to the same position. Unequal keys may describe isomorphic components in different vertex orders. Missing such a merger only loses a possible saving. No graph-isomorphism routine, probabilistic fingerprint, or canonical-form oracle is assumed. Python dictionaries use complete key equality after hashing; a hash collision does not identify unequal components.

5.3 One transition

At each crossing, the implementation performs the following operations on the stored representatives:

1. Attach the crossing and apply the same delooping and exact Gaussian elimination as the ordinary scanner.
2. Track the parent representative of every newly allocated object. Every new differential component must belong to a single parent.
3. Find its connected components, construct their keys, and normalize their degrees.
4. If a child of parent (p, T) has normalization shift h_0 , add $z^{h_0}p(z)$ to the weight of its key. Keep one representative for each key.

The child need not have the same graph as its parent. It may split into many components, vanish after contraction, or remain connected. Multiple children with the same key have their weights added, even when they arose from different parents.

Lemma 5.2 (Extension preserves independent summands). *For the crossing-extension operation F in the $\mathbb{F}_2$ scanner,*

$$F(C \oplus D) \cong F(C) \oplus F(D), \quad F(C[h]) \cong F(C)[h].$$

Furthermore, unit Gaussian elimination can be performed inside each direct summand without changing the homotopy type of the whole complex.

Proof. The crossing complex has two smoothings with a saddle between them. Tensoring and gluing are additive. Each old object produces its own two smoothing blocks, and each transferred old differential remains between objects belonging to the same old summand. The saddle also stays inside the block of one old object. This proves the first identity at the matrix level. Homological shifts commute with extension; the usual signs from a shift are immaterial in characteristic two.

For a unit entry $\phi : C_b^h \rightarrow C_c^{h+1}$, cancellation changes the remaining differential by the Schur-complement term $\gamma\phi^{-1}\delta$. If the matrix is block diagonal by summands, both incoming and outgoing entries of this pivot are in its own block. The correction is therefore in that block. The standard cancellation is a chain homotopy equivalence, so doing it independently in each block preserves the direct-sum homotopy type. $\square$

Theorem 5.3 (Correctness of component sharing). *After every crossing, the complex represented by the compressed state is chain-homotopy equivalent to the corresponding ordinary scanning complex. At final closure, the exact-weight backend returns the same unreduced rank, reduced rank, and raw homological-degree rank counts as the ordinary backend.*

Proof. For $n = 0$, the explicit input convention returns the rank-two unknot complex directly. Assume $n > 0$. Initially there is one monoidal empty matching in degree zero with weight one;

its crossing extensions generate the closed-link complex. Assume the invariant before a crossing. By Lemma 5.2, it is sufficient to extend one copy of each template and replicate the result by the old multiplicities and shifts. Elimination on the representative is a valid contraction for every copy. The resulting differential graph splits as the direct sum of its connected components. Degree normalization records exactly the removed shift, and equality of full keys is a chain isomorphism. The weight update therefore represents precisely the direct sum of all children, proving the inductive step.

The fully closed matching is the empty one. Its endomorphism ring in the implemented algebra is $\mathbb{F}_2$, so a nonzero differential entry is a unit. After complete elimination no differential remains. Each surviving object then contributes one dimension in its recorded homological degree. Expanding the multiplicity polynomials recovers all those dimensions without materializing their generators. The inherited $\mathbb{F}_2$ splitting gives the reduced counts by division by two. $\square$

The proof does not require the compressed scanner and the ordinary scanner to choose the same pivots or to have equal intermediate matrices. Compression rennumbers objects and can change tie-breaking. Both calculations preserve the relevant chain-homotopy type. Tests compare final ranks and degree counts, with $d^2 = 0$ checks as an additional invariant.

5.4 Integer and degree bounds

Lemma 5.4. *After j crossings, the virtual expanded state associated with the compressed run has at most 8^j objects. Every multiplicity coefficient therefore has at most $3j + 1$ bits, and every shift occurring in an exact multiplicity polynomial is between 0 and j .*

Proof. Before extension, every object is a loopless matching. A closed circle newly created by a smoothing must contain at least one of that smoothing's two new arcs. Thus each smoothing creates at most two circles, and delooping creates at most four copies for that smoothing. Both smoothings together create at most eight objects per old one. Gaussian elimination only removes objects. Apply this bound inductively to the virtual expanded process in which the representative's valid contractions are copied on every summand. This process need not coincide with the baseline's pivot sequence.

Each crossing changes the raw cube degree by zero or one. Absolute degrees remain in $[0, j]$. Normalization moves the minimum into the multiplicity shift without changing an object's absolute degree; every template contains an object of relative degree zero. Its shift is therefore also in $[0, j]$. $\square$

The virtual rank can be exponential while its binary length is linear. The exact backend thus avoids an output-size obstruction to listing the basis, while retaining an explicit polynomial-length list of homological ranks. It does not output a basis, cycle representatives, or quantum gradings.

6 Decision-only multiplicities saturated at three

Definition 6.1. *Let $S_3 = \{0, 1, 2, 3\}$ with*

$$a \oplus b = \min(3, a + b), \quad a \otimes b = \min(3, ab).$$

The element 3 means “at least three”. It is not arithmetic modulo three.

Lemma 6.2 (Positive semiring quotient). *The map*

$$q : \mathbb{N}[z] \longrightarrow S_3, \quad q(p) = \min(3, p(1)),$$

is a semiring homomorphism. In particular, $q(z^h p) = q(p)$.

Proof. Evaluation at one preserves addition and multiplication. For nonnegative integers a, b ,

$$\min(3, a + b) = \min(3, \min(3, a) + \min(3, b))$$

and

$$\min(3, ab) = \min(3, \min(3, a) \min(3, b)).$$

For the product identity, a zero factor is immediate; if a positive factor has already reached three, both products saturate. Otherwise neither factor has changed. These identities also give associativity and distributivity by passage from $\mathbb{N}$. $\square$

Theorem 6.3 (Saturated unknot decision). *Replace each exact multiplicity polynomial in the sharing algorithm by its image under q . Retain relative degrees, matching data, and differential entries in each template. Suppose that structural choices and pivot tests do not use the omitted counts or absolute shifts. At completed closure the reported count is exactly*

$$\min(3, r_2(K)).$$

A count of two certifies UNKNOT, and a count of three certifies KNOTTED. Counts zero and one violate the valid-knot promise or an implementation invariant and are not interpreted as knot verdicts.

Proof. At every transition the exact weights change by addition of shifted nonnegative polynomials. Repeated identical children amount to multiplication by a nonnegative integer; more generally all such bookkeeping lies in $\mathbb{N}[z]$. Lemma 6.2 shows inductively that the saturated weights are exactly the images of the exact weights. The same templates are produced because their transitions and equality keys are independent of multiplicities and absolute shifts. Their relative degrees still determine the differential.

At closure, Theorem 5.3 identifies the sum of the exact weights at one with the actual unreduced rank. Applying q gives the stated capped count. The verdict now follows from (2). For a valid knot, the true rank is even, so a capped value of three actually means rank at least four. $\square$

Only the multiplicity storage is saturated. In particular, one cannot cap entries of a differential, signed Euler sums, or intermediate values of a calculation using subtraction under this argument. Two copies of a complex which is later contractible contribute zero at closure, even if their current object count is very large. Thus the mere appearance of a multiplicity three does not authorize early termination.

The decision API reports `rank_capped=3` and `rank_cap=3`; it does not label three as an exact homology rank. It discards absolute grading information by design, so it has no `by_degree` result. Resource exhaustion still raises the scanner's limit exception; the recognition pipeline translates that exception to UNKNOWN.

7 A finite-type complexity theorem

7.1 The actual execution parameters

Fix the scan order and the implemented elimination rule. Let w be the largest number of live boundary labels at any stage of the complete planned order, including the empty frontier. It is even. If Euler inference stops early, unvisited suffix frontiers still enter this definition; a reported visited-prefix maximum need not equal the theorem parameter. Let $b \geq 1$ bound the number of objects in each connected differential component *after that stage's elimination and before merging equal components*. Empty stages may be ignored.

Neither parameter is a topological invariant. In particular, b is not the size of the smallest indecomposable complex obtainable using arbitrary basis changes. It measures components produced by this actual finite algorithm. Observing a small b on some inputs is not a proof of a bound on a family.

Lemma 7.1 (Number of explicitly serialized types). *Set $M(0) = 1$ and $M(w) = (w - 1)!!$ for even positive w . At a fixed stage, the number of different template keys with at most b objects is at most*

$$K(b, w) = \sum_{s=1}^b s^s M(w)^s 2^{s^2 2^{w/2}}. \quad (7)$$

This is an overcount; many counted arrays do not square to zero or do not represent realizable tangles.

Proof. There are at most $M(w)$ pairings of the frontier. Counting all perfect matchings avoids assuming that the scan region has one disc boundary or that numeric edge labels encode a suitable cyclic order. Choose a matching for each of s numbered vertices, in at most $M(w)^s$ ways.

A connected differential graph on s vertices has normalized homological degrees in $\{0, \dots, s - 1\}$. Indeed, any two vertices are joined by a simple path of length at most $s - 1$, and each edge changes degree by one. There are therefore at most s^s degree assignments.

For two matching objects, the overlay has at most $w/2$ circles. In the repository's finite dot-mask model, a morphism has one $\mathbb{F}_2$ coefficient for each subset of those circles. It thus requires at most $2^{w/2}$ bits and has at most $2^{2^{w/2}}$ possible values. Allowing an arbitrary value at each of the s^2 matrix positions gives the final factor. Multiply and sum over s .

No extra permutation factor is required: the vertices were explicitly numbered, so all their ordered serializations are already counted. The implementation can miss isomorphic keys without violating this bound. $\square$

The double exponential inside (7) matters. The quantity $2^{w/2}$ counts coefficient bits, not the number of possible morphisms. Replacing the latter by $2^{w/2}$ would give an unjustifiably stronger theorem. Also, the lemma is about the algebra used by this scanner; it is not a new classification theorem for embedded cobordisms in arbitrary punctured ambient surfaces.

Theorem 7.2 (Complexity with component sharing). *For the finite matching algebra of the existing scanner, both exact-weight and saturated-weight sharing admit the conservative bit-complexity bound*

$$T(n; b, w) \leq n^{O(1)} (bK(b, w))^{O(1)} 2^{O(w)}. \quad (8)$$

In particular,

$$T(n; b, w) \leq n^{O(1)} \exp\left(O\left(b \log(b + 1) + bw \log(w + 1) + b^2 2^{w/2}\right)\right). \quad (9)$$

The same expression gives a conservative space bound. Fixed b, w give polynomial bit complexity, regardless of the virtual multiplicity.

Proof. At any compressed stage there are at most $K(b, w)$ templates and at most $Q = bK(b, w)$ stored objects. Lemma 5.4 bounds the next allocation by $8Q$. Unit elimination uses polynomially many matrix and coefficient operations in this physical size: each pivot removes two objects, and a pivot updates at most quadratically many remaining entries. Maintaining or rescanning candidate lists is polynomial as well.

A dot-mask has $2^{O(w)}$ bits. To compose, enumerate at most $2^{w/2}$ monomials per input, determine each gluing's connectivity by union-find, and expand at most $2^{w/2}$ output decorations. This costs $2^{O(w)}$ bit operations times a polynomial in word lengths; it deliberately allows a less optimized calculation than the compiled implementation. A crossing adds only four local ports. The inherited compiled transfer operations are bounded by the same type of estimate. Graph decomposition, renumbering, and complete key construction are polynomial in the physical matrix and coefficient sizes.

In exact mode, every weight has at most $n + 1$ coefficient positions, each with $O(n)$ bits, by Lemma 5.4. In saturated mode there is one constant-size weight per template. Matching identifiers and input labels require polynomially bounded bit lengths. Across n stages, these observations give (8). It does not rely on average-case hashing: even a quadratic worst-case sequence of key comparisons is polynomial in K and the key length.

Finally,

$$K(b, w) \leq b^b M(w) b^{2^{w/2}}, \quad \log M(w) = O(w \log(w + 1)).$$

Substitution gives (9). Persistent caches, including matching interning across stages, add at most polynomial factors in n and the same size parameters; they do not require expanding multiplicities. $\square$

Corollary 7.3 (A specific quasi-polynomial regime). *If a family admits scan orders for which the actual shared runs satisfy*

$$b^2 2^{w/2} = O((\log(n + 2))^2), \tag{10}$$

then exact homological ranks, and in particular unknot decisions, are computable along those runs in $n^{O(\log n)}$ bit operations. This statement includes the cost of multiplicity arithmetic.

Proof. For integer $b \geq 1$ and even $w \geq 0$, the terms $b \log(b + 1)$ and $bw \log(w + 1)$ are bounded by a universal constant times $b^2 2^{w/2}$. Apply Theorem 7.2. A polynomial-time construction of the promised scan order can be included without changing the conclusion. Alternatively, the statement applies to an order supplied as part of the input. It does not assert that searching all orders is inexpensive. $\square$

Examples of sufficient regimes are $w = O(1)$ with $b = O(\log n)$, and $b = O(1)$ with $w \leq 4 \log_2 \log(n + 2) + O(1)$. A condition such as $b = (\log n)^a$, $w = c \log_2 \log n + O(1)$ gives an exponential bound with logarithmic exponent at most $2a + c/2$, up to the displayed polynomial factor. It reaches the specific target (1) when $2a + c/2 \leq 2$. More generally it gives some quasi-polynomial bound for fixed a, c .

7.2 What the theorem removes, and what it leaves

The theorem contains no upper bound on the number of repeated copies of one template. That quantity may be $2^{\Theta(n)}$. It replaces a bound on all explicit generators by a bound on the number of possible small components in a finite boundary algebra.

There is no proved universal estimate (10). Large connected components and large frontiers remain genuine obstructions. The safe sufficient condition is deliberately conservative; it counts all morphism masks and all matrices. A useful family may run rapidly even when this worst-case estimate is enormous. Conversely, a small observed frontier alone does not make the theorem apply.

One must also distinguish an observed successful execution from a family algorithm. A theorem about a family needs a way to construct a suitable order and prove the component-size bound throughout every run. The measured values included in the artifact package are diagnostic data for this research problem, not substitutes for that proof.

8 Early decisions from componentwise Euler continuations

8.1 Why a continuation is needed

Saturation alone permits an exact decision at closure. To stop earlier, we must prove that enough of the current virtual state survives all remaining crossings. A large current multiplicity does not do that. An exact Euler calculation of each *completed independent summand* supplies a safe lower bound, without computing its full homology.

Let F_i denote extension through the unprocessed suffix, followed by closure. It is the same additive, homotopy-preserving operation used by the ordinary scan. For a compressed state after i crossings, define

$$B_i = \sum_a p_a(1) |\chi(F_i(T_a))|. \quad (11)$$

Absolute values are taken separately for the different templates before combining their multiplicities. Using $p_a(-1)$ would allow copies in different homological shifts to cancel and would lose this information.

Theorem 8.1 (Monotone lower bound for final rank). *Along a complete exact component-sharing run,*

$$B_i \leq r_2(K), \quad B_i \leq B_{i+1}, \quad B_n = r_2(K).$$

Thus $B_i \geq 3$ certifies KNOTTED before closure. The cap $\min(3, B_i)$ can be computed from saturated multiplicities, provided the signed Euler characteristics themselves are computed exactly.

Proof. The direct-sum representation and preservation of homotopy equivalence give

$$r_2(K) = \sum_a p_a(1) \dim_{\mathbb{F}_2} H(F_i(T_a)).$$

For any finite-dimensional complex C , the Euler identity and triangle inequality give $|\chi(C)| \leq \dim H(C)$. Applying this separately to the completed templates proves the lower bound.

Suppose extension and reduction of one template produces $E_i(T_a) \simeq \bigoplus_u U_u[e_u]$. Compatibility of continuation gives

$$\chi(F_i(T_a)) = \sum_u (-1)^{e_u} \chi(F_{i+1}(U_u)).$$

The triangle inequality, multiplied by $p_a(1)$ and summed over parents, proves monotonicity. Merging equal normalized templates adds nonnegative multiplicities and does not change the sum of absolute values. At final closure, the fully eliminated complex is a direct sum of isolated objects; every normalized template has Euler characteristic one. Hence B_n is the actual total rank.

Finally, multiplication by $|\chi|$ and summation are nonnegative operations. Lemma 6.2 shows that saturated multiplicities suffice for the cap of (11). The threshold follows from (2). $\square$

This is an obstruction to triviality. A value $B_i = 2$ at an incomplete stage is inconclusive. Some completed summands have nonzero homology and Euler characteristic zero, so even genuine splitting need not produce an early certificate.

8.2 An exact suffix dynamic program

For a matching m on the frontier after crossing i , set $E_i(m) = \chi(F_i(m))$. The final empty matching has $E_n(\emptyset) = 1$. This is the monoidal empty object; the separate API convention that an empty *input PD* is one unknot is handled separately.

Suppose the next crossing gives matching m_0 and c_0 closed circles in smoothing zero, and matching m_1 and c_1 closed circles in smoothing one. The two cube terms differ in degree by one, while each delooped circle has two generators in that same homological degree. Therefore

$$E_i(m) = 2^{c_0} E_{i+1}(m_0) - 2^{c_1} E_{i+1}(m_1). \quad (12)$$

For a normalized template,

$$\chi(F_i(T)) = \sum_{v \in T} (-1)^{h_v} E_i(m_v). \quad (13)$$

The differential of T is unnecessary for this particular calculation, because Euler characteristic depends only on chain dimensions.

The implementation memoizes pairs consisting of the suffix index and the full matching. It evaluates the dependency DAG using an explicit stack, so a long narrow braid does not hit Python's recursion limit. It deliberately uses (12) rather than a writhe shortcut. The recurrence follows the scanner's exact raw degree convention and works for arbitrary valid scan permutations without additional orientation formulas.

Proposition 8.2 (Cost of the Euler continuation). *If all frontiers have size at most w , the complete memo table has at most $(n + 1)M(w)$ states. Each value has $O(n)$ bits. With ordinary finite-state dynamic programming, all requested continuation values cost $n^{O(1)}2^{O(w \log(w+1))}$ bit operations. Combining them with the physical templates adds a polynomial cost in their stored size.*

Proof. The boundary label set at each stage is fixed by the scan order, independently of smoothing choices. There are at most $M(w)$ complete matchings on it. Each state has two outgoing dependencies. From (12) and $c_0, c_1 \leq 2$, induction gives $|E_i(m)| \leq 8^{n-i}$, so the bit bound is linear. The two gluing operations are polynomial in w ; memoized integer addition and multiplication by powers of two have polynomial bit cost. Use $M(w) = 2^{O(w \log(w+1))}$. Full key equality instead of constant-time hashing can be charged by a further polynomial factor in the state count, which gives the same displayed exponential form. $\square$

The coefficient values in (12) must not be capped. Subtraction may cancel two large terms to a small integer. Only after an exact signed sum and an absolute value may the nonnegative lower bound be saturated. This is tested explicitly, including mirror diagrams and intermediate values larger than the final Euler magnitude.

The optional implementation uses a state budget, defaulting to 4096. If the budget is exhausted, it disables only this inference and continues the complete saturated scanner. A time or physical-object limit remains a limit on the whole computation and becomes UNKNOWN. The early result exposes `rank_lower_bound_capped`, not `rank` or `rank_capped`; the latter would misleadingly suggest a completed homology calculation. The suffix-state budget includes all registered states over the lifetime of the inference engine.

8.3 Scope of the early-decision result

The theorem supplies a new proof obligation that can be checked without assuming that current objects survive: calculate the Euler characteristic *after the actual remaining continuation*. The bound is stronger than the absolute Euler characteristic of the whole complex because opposite Euler contributions from independent components are not cancelled.

It is still sensitive to the differential-component decomposition reached by the scan. If there is one component with multiplicity one, its completed Euler magnitude is just the ordinary knot Euler magnitude. The implementation therefore waits for an actual split before attempting the optional dynamic program. This is an optimization of a sound condition, not a completeness assumption. An unbounded suffix calculation fits within the conservative finite-type bound of Section 7; the budgeted version gives better control of practical overhead.

9 Implementation, validation, and measured effects

9.1 Integration choices

The archive contains an integrated **fastunknot** version 0.3.0 in **fast/**. The unchanged pinned source is under **reference/fast/**; the supplied patch applies the implementation changes to the original repository path. No remote branch or commit is modified by this deliverable.

The structural certificate runs before the legacy reduction and filtering pipeline. This placement gives linear recognition on homogeneous input diagrams without first allocating an Alexander matrix. It also imposes an extra traversal when inconclusive, including on a diagram that elementary reduction would immediately settle. The option `use_seifert=False`, or `-no-seifert` on the command line, restores the previous stage ordering. This is a measured tradeoff, not a universal speedup claim.

The ordinary scanner remains the default fallback. The new selectable backends are **shared**, **saturated**, and **euler**:

Backend	Information retained	Returned homology evidence
standard	Original explicit representatives	Exact ranks and raw homological-degree counts
shared	One representative per exact key; $\mathbb{N}[z]$ weights	The same exact ranks and degree counts
saturated	Relative template degrees and S_3 weights	Final rank_capped , with cap three
euler	Saturated sharing plus bounded exact suffix evaluation	An early rank_lower_bound_capped , or a final rank_capped

The new backends currently use the existing bit algebra and minimum-fill pivot policy. Incompatible requests for the set algebra, alternative pivots, tail contraction, or multi-process scan races are rejected explicitly. They are not silently ignored. The exact-rank command also accepts `-shared`. The low-level interfaces retain the valid classical one-component input promise described in Section 2.

In shared modes, **max_objects** limits actual allocated representative objects before elimination. It does not limit the number of virtual copies. Time limits are cooperative rather than hard process deadlines, as in the baseline. Object or time exhaustion gives UNKNOWN. Exceeding the optional Euler-state budget only disables that inference and continues the exact decision scan. Early Euler evidence is a certified lower bound and is never labeled as a completed exact rank. In saturated mode, statistics concerning expanded multiplicity are named as lower bounds.

9.2 Validation design

The final test log accompanies the code. The suite has 69 test methods: 41 inherited methods, six structural-certificate methods, nine component methods, eight Euler methods, and five public integration methods. The two inherited pipeline test modules explicitly disable the new structural stage when asserting old method names; new tests exercise the new default. This retains the meaning of the baseline regression tests rather than changing their expected intermediate methods wholesale.

The checks address independent failure mechanisms:

- Structural data are compared with closed-braid formulas on 160 fixed-seed random closures, including mirrors, row half-turns, crossing permutations, and edge relabelings. Issued certificates are compared with exact Khovanov recognition on 60 short random closures. An independent orientation reconstruction and union-find verifier replays certificates; forged fields and the misleading zero Rasmussen interval are tested.
- Component sharing is compared with the original scanner on 200 fixed-seed random valid closures, half with arbitrary random crossing orders. The suite checks $d^2 = 0$ after every scan stage and includes comparisons with the independently represented generic set algebra. Tests cover exact-key equality, integer rather than parity multiplicities, degree shifts, mirrors, torus examples, and resource limits.
- Suffix Euler values are compared with explicit continuation using the generic algebra on 40 random prefixes. Fifty random arbitrary-order diagrams check monotonicity of the bound and equality with the rank at final closure. Additional tests cover signed cancellation, mirror conventions, safe budget exhaustion, and a 1,201-crossing order handled without recursion.
- Integration checks compare the default pipeline with the legacy pipeline on 100 random closures, exercise all four backends, replay default certificates, and check JSON fields and command-line rejection of incompatible options.

The finite family check in `benchmarks/defect_one_family.py` independently tests the symbolic formulas of Section 3.5 for all $2 \leq k \leq 100$: 99 diagrams through 202 crossings. This verifies the implementation of the family; the proof for arbitrary k is the symbolic argument in the article. These tests support an exact implementation but are not a machine-checked proof of the entire inherited cobordism algebra or of the external topological theorems.

9.3 Structural-stage experiment

Twenty cases were measured in nine randomized, interleaved rounds. Each round included the baseline arm A , the new structural-first arm B , and two identical baseline control arms. Every invocation received a fresh, uncached *already validated* PD diagram. The measured boundary is the recognition function; parsing, validation, imports, file I/O, and result serialization are excluded symmetrically. This is the boundary at which the front end is inserted. Both implementations share the same unchanged fallback pipeline.

For each case, paired speedup is the geometric mean of the nine A/B wall-time ratios. The intervals below are 95% percentile bootstrap intervals of the paired mean log ratio, with 3,000 fixed-seed resamples. They describe variation in this shared-host experiment, not a universal hardware-independent confidence statement. The median times and paired ratio are different summaries and need not divide to the same value.

The baseline decided these cases with its modular Alexander stage. The new certificate avoids that stage, including its dense matrix allocation. The mathematical improvement is an $O(n)$ word-

Family	n	Median A (ms)	Median B (ms)	Paired speedup
$(\widehat{\sigma_1\sigma_2^{-1}})^{500}$	1,000	62.1	2.14	28.75
$(\widehat{\sigma_1\sigma_2})^{500}$	1,000	60.6	2.05	25.97
$(\widehat{\sigma_1\sigma_2\sigma_3\sigma_4})^{251}$	1,004	129.8	2.02	53.94

Table 1: Recognition-function measurements on homogeneous families. The paired speedup intervals are respectively [25.59, 31.98], [20.97, 32.83], and [44.86, 63.75]. All rows are knots.

operation decision for this class; the timing data illustrate that improvement but do not estimate the exponent of the general recognizer. The usual $O(n^3)$ field-operation bound on dense elimination is an upper bound, and these timings do not claim that the chosen braid family attains it.

A second complete experiment pinned execution to CPU 8, disabled cyclic garbage collection, and used longer timing batches. Its paired speedups on the same three large cases were 27.86, 28.88, and 44.13. The large-family advantage therefore persisted under this changed measurement protocol. The original and repeated raw observations are both included.

Inconclusive controls expose the cost of the additional pass:

Control	Original paired A/B	Repeated paired A/B
Conway knot	0.861	1.003
Kinoshita–Terasaka knot	0.918	0.883
Eight-crossing hard unknot	0.770	0.953
Six-crossing reducible grid unknot	0.478	0.470

Ratios below one favor the baseline. The tiny grid example rises from about 22 microseconds to 49–52 microseconds. Identical-arm controls show substantial noise on tiny cases, even after pinning; moderate percentage changes on the other controls should not be read as precise stable effects. The substantial large-family gains and the approximately 30-microsecond cost on the immediately reducible grid input are the robust conclusions.

9.4 Component-sharing experiment

The scanner experiment deliberately disables geometric factorization and all recognition filters. It measures the algebraic backend itself. Seven shuffled rounds per case contain two original-scanner arms, full sharing, and saturated sharing. The recorded crossing order and output ranks are checked. Define the deterministic work counter as

$$W = \text{entries} + \text{compositions}.$$

This counter omits Python bookkeeping, key construction, multiplicity updates, and Euler work. It is useful for interpreting the algebraic change; it is not a machine-independent running-time bound.

Saturated-sharing time ratios on the same rows are 1.188, 1.145, 1.284, 0.253, and 0.00793. The identical-arm ratios range approximately from 0.85 to 1.34 across these runs. The repeated-sum savings greatly exceed that variation. On the prime controls, sharing does not reveal a useful decomposition until very late and adds overhead. Different pivot ordering after representative normalization can also change algebraic work; the stress example’s work increases slightly. Hence sharing is an optional backend, with no claim that it dominates the existing scanner.

Eight Conway summands provide a larger representation check. The exact unreduced rank is

$$2 \cdot 33^8 = 2,812,817,236,482.$$

Input	W old	W shared	Peak old	Peak shared	Full time ratio
Conway	1,609	1,605	246	246	1.220
$T(3, 5)$	397	397	54	54	1.220
Stress braid, 36 crossings	275,165	285,445	17,694	17,694	1.312
Two Conway summands	58,844	6,875	8,118	975	0.244
Three Conway summands	1,952,418	12,437	267,894	975	0.00735

Table 2: Raw scanner comparison. Peak means objects allocated before elimination. Time ratio is median paired new/old, using the geometric mean of the two old arms within each round. Smaller ratios favor sharing.

The sharing run uses at most 975 objects before elimination and 240 representative objects after sharing. The full mode performs 16,162 multiplicity updates; saturation reduces that counter to 377. A single diagnostic run completed in about 0.095 seconds in full mode and 0.088 seconds in saturated mode. These single timings are not a paired speedup estimate; the exact rank, allocation counts, and update counts are the reproducible evidence of compression.

This repeated-sum experiment does not establish an end-to-end recognition improvement. The pinned recognizer already splits visible connected sums geometrically and can reject one nontrivial factor. It also has polynomial filters that settle many of the prime controls. The experiment establishes that literal repeated homological summands can be processed without expanding their multiplicities. Its contribution to general recognition remains dependent on finding useful analogous structure in the filter-resistant prime residual class.

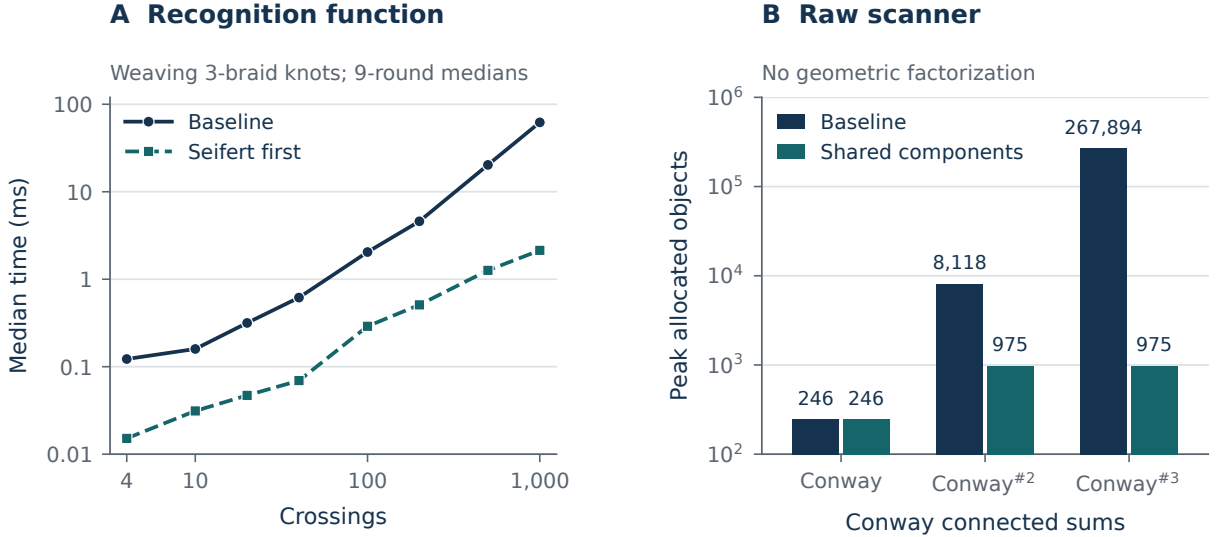


Figure 1: Two improvements with different measurement boundaries. (A) Median recognition-function times on weaving three-strand knots in the original structural experiment, nine rounds per input; parsing and validation are excluded. Logarithmic axes and connecting segments show measurements, with no fitted exponent. (B) Recorded maximum objects before elimination in the raw baseline and exact shared scanners on one, two, and three Conway summands, with geometric factorization disabled. These are scanner-object counts, not process-memory measurements.

9.5 Early Euler certificates

The exact continuation recorder gives the following operation counts:

Input	Decision stage	Euler states	Scan W at decision	Full shared W
Two Conway summands	11/22	34	5,252	6,875
Three Conway summands	11/33	76	5,407	12,437
Eight Conway summands	11/88	286	5,252	39,317

In each certificate the retained template has exact completed Euler characteristic +2 and saturated multiplicity three. The resulting nonnegative rank lower bound is capped at three and already exceeds the unknot threshold. A one-state Euler budget on the eight-summand input exhausts safely and completes the ordinary saturated scan at stage 88.

The Conway knot, stress braid, and hard-unknot controls trigger no Euler states because they do not split early. The $T(3, 5)$ example uses three Euler states but yields no early verdict. It is a useful reminder that nontrivial homology can have zero Euler characteristic within a summand. The table counts Euler states separately: omitting their cost from W must not be interpreted as omitting it from running time. No general Euler-engine speedup is claimed from these diagnostic observations.

9.6 What the evidence does and does not establish

The structural front end changes the asymptotic cost of recognition on a large, mathematically defined diagram class. The shared representations remove an explicit multiplicity barrier and satisfy the finite-type theorem. The Euler extension can use surviving summands as exact early witnesses. All three are implemented and reproducibly tested.

The observed examples do not bound the sizes of future connected components, prove a small frontier for arbitrary inputs, or certify that random tests represent hard unknot diagrams. No fitted empirical exponent substitutes for those missing structural statements. The default remains a complete recognizer with an exponential worst-case upper bound, with UNKNOWN under optional resource ceilings. The following research programme addresses the missing general theorem directly.

10 The remaining route to general quasi-polynomial time

10.1 Two routes with different missing theorems

The current implementation can now avoid work in three distinct ways: recognize a certified geometric class immediately; share repeated direct summands; or prove sufficient final homology dimension from a partial scan. None of these mechanisms gives a universal estimate on the frontier or on the connected differential components.

The algebraic route needs a theorem that constructs a useful decomposition of *every* knot diagram, with controlled component complexity under continuation. The hierarchy route needs an executable representation of patterned handle structures, compressed surface operations, and a global bound on effective restarts. These are different programmes. A successful subroutine for one programme does not automatically discharge the other programme's complexity obligations.

For the algebraic route, a quantitative target is to prove for some constructively specified ordering or more general assembly rule that

$$b^2 2^{w/2} = O(\log^2 n).$$

This target is sufficient rather than necessary. The type-count proof is crude enough that a much sharper classification of reachable morphisms or templates could replace it. It may be more realistic to control the number of *reachable* templates directly than to force every template to have logarithmic size.

There is also a topological obstruction to reducing the frontier on every input merely by choosing a better crossing order. The tree-width theorem of de Mesmay, Purcell, Schleimer, and Sedgwick implies that every diagram of a torus knot $T(p, q)$ has tree-width at least $\min(p, q)/4 - 1$ [6, Theorem 3.7]. A linear frontier scan induces a carving decomposition up to constant factors, so the standard $n = \Theta(p^2)$ diagrams of $T(p, p+1)$ require $w = \Omega(\sqrt{n})$ under every such ordering. Their homogeneous structure is already handled by the new front end. A useful general width theorem must therefore concern the residual class after certificates, or a genuinely different representation. Width of a knot diagram and width of a triangulated knot complement must not be conflated; the same source shows that torus-knot complements admit constant-tree-width triangulations.

10.2 Nonuniform restart accounting for a future hierarchy engine

The following elementary result is useful when different hierarchy levels have substantially different complexity caps. It is stated independently of any claim that the necessary geometric invariants have been implemented.

Proposition 10.1 (Mixed-radix restart bound). *Suppose an algorithm has a length- L state vector $(a_1, \dots, a_L)$ with integer bounds $0 \leq a_i \leq g_i$. At each restart, the first changed coordinate strictly decreases; all subsequent coordinates may reset arbitrarily within their fixed bounds. Then the number of such restarts is at most*

$$\prod_{i=1}^L (g_i + 1) - 1.$$

If every between-restart operation has polynomial bit cost and all stored encodings have polynomial size, a sufficient condition for the target $n^{O(\log n)}$ is

$$\sum_{i=1}^L \log(g_i + 1) = O(\log^2 n),$$

with polynomially many operations between successive restarts.

Proof. Use the mixed-radix potential

$$\Phi(a_1, \dots, a_L) = \sum_{i=1}^L a_i \prod_{j>i} (g_j + 1).$$

It is an integer between zero and $\prod_i (g_i + 1) - 1$. If coordinate i is the first to change, decreasing it by one lowers Φ by $\prod_{j>i} (g_j + 1)$. The entire suffix can increase by at most that quantity minus one. Every restart therefore decreases Φ by at least one. Taking logarithms of the resulting product bound gives the last statement after charging the polynomial overhead. $\square$

The uniform case with $g_i = g$ gives the familiar $(g + 1)^L$ factor. Polynomial surface caps with $L = O(n)$ still permit $n^{O(n)}$ restarts. Polynomial caps with $L = O(\log n)$ yield (1); with $L = O((\log n)^2)$ they give $2^{O((\log n)^3)}$. Coordinate compression alone does not change this arithmetic.

The proposition is a template for a proof, not an observation that the geometric algorithm already has this potential. The caps must be globally valid after every reconstruction. A log recording decreasing numbers does not establish either that validity or the topological correctness of the restart operation.

10.3 Concrete interface obligations for a hierarchy implementation

The minimum coherent implementation unit is larger than a normal-coordinate vector or a ball-pattern tester. It should include the following contracts.

1. **Encoded topology.** A finite list of allowed handle types, exact attachment maps, boundary-pattern incidences, orientation, and an encoding size charged to the input and all created objects.
2. **Compressed surface operations.** Binary multiplicities, component and boundary information, cutting, gluing, and transport of curves or discs across each change. A procedure must operate on binary size, rather than expand one disc per unit of normal weight.
3. **Topology-preserving simplification.** Every parallelity-bundle replacement or handle cleanup must carry a precise statement of the manifold and boundary pattern it preserves or changes.
4. **Progress.** Surface production and violating-disc transport must support a globally bounded potential through every replacement of a hierarchy suffix, including changes to the representation itself.
5. **Cost composition.** Bound the number of generated handles, coefficient lengths, calls, and restarts together. A fast primitive is insufficient if called super-quasi-polynomially often.

The Agol–Hass–Thurston orbit-count method is an established model for processing exponentially long interval systems through short encodings [2]. Its existence supports this design direction; it does not, by itself, implement the full contracts above. The present archive leaves these interfaces as a research programme rather than supplying placeholders that report topological verdicts.

11 Further research questions and proposed work

The following questions are ordered by the strength and immediacy of the connection to the delivered implementation. Each is paired with a concrete success criterion.

- Q1. Which prime families have bounded component size after sharing?** Find an infinite family of prime, nonhomogeneous diagrams for which the actual component-sharing policy has a proved bound on b and w , and for which earlier filters are not the entire explanation of fast recognition. A proof must specify the scan order and the representative transition types. A list of observed small components is useful evidence but not the result.
- Q2. Can reachable morphisms replace the all-masks bound?** The factor $2^{s^2 2^{w/2}}$ counts every matrix entry in the full dot-mask space. Identify a closure property of the entries actually produced by extension and unit elimination. A bound on the number of reachable coefficient masks could enlarge the quasi-polynomial regime substantially. The proposed class must survive composition and Schur-complement addition.
- Q3. Can connected components be split efficiently after a basis change?** The current graph decomposition finds literal blocks. Develop a restricted, provably polynomial family of basis changes that reveals further repeated summands. The output should include an explicit invertible change of basis or chain-homotopy equivalence. A general canonical-form oracle would merely move the hard computation elsewhere.

- Q4. How much can a pointed or reduced category expose?** A carefully constructed marked-edge implementation might kill some nonunit morphisms earlier and reveal smaller components. It must track the basepoint through all gluing and delooping operations, prove compatibility with reduced homology, and preserve a finite composition algebra. This archive does not include an unverified quotient obtained by simply deleting dot terms touching a chosen boundary point.
- Q5. Can continuation-aware obstructions be strengthened?** Theorem 8.1 uses one scalar Euler functional for each completed summand. Investigate several quantum-graded evaluations or other additive functionals with certified norm bounds. The central requirement is a proved lower bound on homology rank after continuation. Capping signed sums before cancellation would invalidate such a certificate.
- Q6. When should the optional Euler engine run?** Develop a predictor based on cheap component statistics or a small number of exact suffix values. Measure total time including failed inference. Any adaptive policy must preserve the property that a budget failure simply returns to a complete backend. Establishing a profitable policy on prime filter-resistant diagrams would be more informative than optimizing the visible-sum stress family again.
- Q7. Can a complete three-braid rank engine be integrated?** The polynomial-time three-braid result in the 2026 literature suggests a specialized exact backend based on normal forms and formulas [7]. Implementation requires careful grading, mirror, linking, and coefficient conventions. Validate it against the independent cube and both scanning algebras before using it to bypass scanning. The mere fact that the input has three strands is not a complexity theorem for the current scanner.
- Q8. Can extremal-band complexity be made uniform?** Seek a fully quantified finite-field bound such as $2^{O(w)}n^{O(k)}$ for computing a band of k homological degrees under an explicitly specified assembly rule. The number of objects, decorated morphisms, coefficient sizes, and the extra degree needed for outgoing differentials must all be charged. A complete recognizer would still need a proved location of a nontriviality witness in those bands or a fallback.
- Q9. What structural parameter survives reduction?** Example 3.4 and the I/II-reduced family in Section 3.5 rule out control by homogeneity defect alone. Search for a parameter combining signed block structure with a certified diagram reduction or tangle decomposition. It should distinguish many redundant twists from genuine mixed-sign complexity.
- Q10. Can a controlled portfolio improve the heavy tail?** Sharing, saturated sharing, Euler inference, the ordinary scanner, and geometric reduction have different overheads. A portfolio should be judged by CPU work as well as wall time. A useful guarantee would compare its cost with the best applicable complete backend, while bounding the work lost in abandoned runs. Purely retrospective selection of the fastest run is not an implementable performance claim.
- Q11. Can the new proof obligations be formalized?** A small formal development can first prove the S_3 semiring quotient, weighted direct-sum transition invariant, and monotone Euler bound. These have explicit finite algebraic interfaces. Formalizing the full Bar-Natan category and the unknot-detection theorem is a separate, substantially larger task. A checked combinatorial certificate still depends on its imported topological soundness theorem.
- Q12. Can the hierarchy route obtain a nonuniform progress bound?** Implement a complete unaccelerated hierarchy engine with explicit contracts, then search for level-dependent caps

satisfying Proposition 10.1. A successful result would specify the caps, the retained suffix under each simplification, and the proof that every expensive restart is counted. This is a direct path to a real end-to-end quasi-polynomial theorem, but it remains beyond the delivered changes.

12 Concluding assessment

The delivered recognizer has a broader polynomially decidable front end and an exact mechanism for representing repeated chain-complex summands without expanding their generators. Saturated weights specialize that representation to the actual yes/no problem, and exact Euler continuations supply a safe form of early nontriviality detection. The finite-type theorem identifies a specific sufficient regime for $n^{O(\log n)}$ time that accounts for both algebraic state size and integer arithmetic.

The implementation retains an exponential worst-case upper bound; no general quasi-polynomial bound is proved. The measured improvement on homogeneous families is an end-to-end pipeline improvement after validation. The dramatic connected-sum scanner figures are representation stress tests, because an earlier specialized factorizer already handles those examples efficiently. The hardest remaining task is to control structural complexity for diagrams that are neither certified by the front end nor decomposed into small repeated components. The archive provides an executable basis for pursuing that task and exact criteria for judging whether a future claim has actually reached the quasi-polynomial target.

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